Requirements

ISE Coursework — Bug Report Classification

1 Python Version

Python 3.9 or higher. Python 3.11 is recommended. To check your version:

```
python3 --version
```

2 Required Packages

All dependencies are standard scientific Python libraries. The table below lists what is needed and why.

Package	Version	What it is used for
scikit-learn	1.3 +	TF-IDF vectoriser, Naive Bayes, Random Forest, evaluation metrics
scipy	1.11 +	Wilcoxon signed-rank test for statistical comparison
numpy	1.24 +	Numerical operations and array handling
pandas	2.0 +	Reading CSV files and writing results

3 Installation

Install everything in one command:

```
pip install scikit-learn>=1.3 scipy>=1.11 numpy>=1.24 pandas>=2.0
```

4 Operating System

Tested on Windows 10/11, Ubuntu 22.04, and macOS 13+. Any OS that runs Python 3.9+ should work fine.

5 Hardware

No GPU needed. A standard laptop with 4 GB RAM is more than enough. The full 5-project, 30-run experiment takes around 3–8 minutes on a modern CPU. The Random Forest uses all available cores by default (n_jobs=-1) so more cores means faster runs.

6 Dataset

Download the five project CSV files from the lab repository:

```
https://github.com/ideas-labo/ISE/tree/main/lab1
```

Place them in a folder called `data/` in the same directory as the script. The expected filenames are: TensorFlow.csv, PyTorch.csv, Keras.csv, MXNet.csv, Caffe.csv.